

LAB ASSIGNMENT

Q-1 Decision Tree for Heart Disease Prediction

Part A – Basic Implementation

- Train a Decision Tree on the Heart Disease dataset.
- Report training and testing accuracy.
- Visualize the decision tree and interpret the first two splits.

Part B – Model Evaluation

- Print the **confusion matrix** and **classification report** (precision, recall, F1-score).
- Plot the **ROC curve** and report the **AUC score**.

Part C – Hyperparameter Tuning

- Vary max_depth, min_samples_split, min_samples_leaf.
- Compare training vs testing accuracy → shows **overfitting vs underfitting**.
- Use **GridSearchCV** for automated hyperparameter tuning.

Part D – Error Analysis

- Find the patients (rows) that were misclassified by the decision tree.
- Compare their features with correctly classified patients.
- What patterns do you notice? (e.g., “younger patients with mild symptoms were misclassified”).

Part E – Class Imbalance Check

- Check class distribution (how many patients with disease vs without).
- If imbalanced → use metrics like **balanced accuracy** or apply resampling (SMOTE, undersampling).
